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***B.Tech. Degree II Semester Regular and Supplementary/
I Semester Supplementary Examination April 2018***

**CE/EE/ME/SE GE 15 – 1201A & CS/EC/IT GE 15 – 1101 B
COMPUTER PROGRAMMING
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A*(Answer ALL questions)**(10 × 2 = 20)*

- I. (a) What is system software?
- (b) List the functions of loader and linker.
- (c) Explain the built in data types available in C.
- (d) Explain the different branching statements.
- (e) Find the value of Monday and Wednesday in the following statement. enum day {Sunday=0, Monday, Tuesday=4, Wednesday}. Give the explanation of your answer.
- (f) Write a program to generate a set of random numbers.
- (g) Write a program to check whether a substring is present in given string.
- (h) Compare array of pointers and pointer to an array.
- (i) Explain c-alloc () and m-alloc () functions.
- (j) Briefly describe the dynamic memory allocation.

PART B*(4 × 10 = 40)*

- II. Describe the building block of a digital computer system. (10)
- OR**
- III. Explain the different design tools with examples. (10)
- IV. Explain the different control statements in 'C'. Give example for each. (10)
- OR**
- V. (a) Write a program to find the GCD of 2 numbers. (5)
- (b) Write a program to print the prime numbers between 0 and 100. (5)
- VI. What is bubble sorting? Write a program to sort a list of names in alphabetic order using bubble sorting. (10)
- OR**
- VII. (a) How are functions declared? Give an example of a recursive functions. (5)
- (b) Explain the user-defined data types. (5)
- VIII. (a) Explain all file handling operations. (4)
- (b) Write a program to find the product of 2 matrices using pointers. (6)
- OR**
- IX. (a) Read 2 numbers from command prompt and find the sum of these numbers using command line arguments. (5)
- (b) Write a program to open a file and change all characters to lower case. (5)

